

Can Humans Detect AI?

Mining Textual Signals of AI-Assisted Writing Under Varying Scrutiny Conditions

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Abstract

This study asks whether the *threat* of AI detection changes how people write with AI, and whether other people can tell the difference. I built a two-phase controlled experiment where 21 participants wrote opinion pieces on remote work using an AI chatbot. Half were randomly warned that their submission would be scanned by an AI detection tool. The other half received no warning. Both groups had access to the same chatbot. In Phase 2, 251 independent judges evaluated 1,999 paired comparisons, each time choosing which document in the pair was “written by a human.” Judges were not told that both writers had access to AI. Across all evaluations, judges selected the warned writer’s document as human 54.13% of the time versus 45.87% for the unwarned writer. A two sided binomial test rejects chance guessing at $p = 0.000243$, and the result holds across both writing stances. Yet on every measurable text feature I extracted, including AI overlap scores, lexical diversity, sentence structure, and pronoun usage, the two groups were indistinguishable. The judges are picking up on something that feature based methods do not capture.

1 Introduction

Large language models have made AI-assisted writing nearly indistinguishable from human-authored text. This has raised real concerns in academia, journalism, and the workplace about whether anyone is still writing their own content or relying on AI tools to write content for them. Universities for example have responded by adopting AI detection tools. The question covered in this paper will look at whether or not a *threat* of AI-detection actually changes how participants write with AI, and if it does, can other people tell?

This project centers the question through a two-phase controlled experiment. I built a Streamlit app that sets up participants with an AI chatbot and randomly assigns them to one of two conditions. The control group receives no extra context, while the treatment group is told that their final submission will be analyzed by an “AI detection tool.” Everyone answers the same prompt to form a pair of opinion pieces on remote work, one “For” and one “Against.” In Phase 2, a separate pool of judges reads the paired documents (one control, one test, same stance) and has to pick which one they believe was “written by a human.” They go through a funneled survey designed in Streamlit that allows them to choose which of our writer’s responses were “written by a human.” The judges’ task was intentionally deceptive. Both groups in the experiment had access to identical chatbots, so judges were not picking “human vs AI.” Instead, the response from the judges was testing to see if the judge can identify which writer had been warned that detection software would read the result. The initial hypothesis of this research was that writers who were warned of an AI detection

software were more likely to edit and re-frame their writing to sound more “human,” and judges would have been able to identify writings in the test group at a significantly higher rate as humans compared to the control group. The hypothesis was intended to measure the incremental behavioral change of an AI-detection threat for the treatment group.

The two research questions are:

1. **RQ1 (Behavior under scrutiny):** Do writers behave differently when they know their submission will be scanned for AI? This covers both interaction metrics (turns, duration, tokens, self-reported reliance) and textual features of what they submit.
2. **RQ2 (Human judges):** Do human judges detect AI reliance? Does confidence, reading time, or belief alignment moderate their accuracy?

Preview of findings: The initial hypothesis was directionally supported, but the treatment effect is small. Across $n = 1,999$ paired evaluations from 251 judges, the warned (test) document was identified as the human written one 54.13% of the time, versus 45.87% for the unwarned (control) document (95% Wilson confidence interval on the control as human rate: [43.70%, 48.06%]). A two sided exact binomial test against the chance rate of 0.5, treating each response as independent, rejects random guessing ($p = 0.000243$ which is significant at $\alpha < 0.05$). Judges on average rated the warned writers as more human, which is consistent with the hypothesis, but the difference is only a few percentage points above chance. However, this difference is not insignificant.

The more interesting finding is the disconnect between that signal and the writing itself. On every behavioral and style feature tested, warned and unwarned writers were indistinguishable. Both groups leaned on the chatbot at similar rates: control writers had an AI overlap score of 0.978, test writers 0.953. The overlap score is an engineered score (0 to 1 scale) that measures how much of a submission matches word for word any AI response the writer saw during the chat in their session (It doesn’t consider turns or draft edits - it only looks at how much of the submitted text matched a full AI response in the chat). 1 means the submission is an exact copy-paste from an AI message, and a score near 0 means the writer rewrote the text in their own words or never worked with the chatbot to begin with. The score catches copy-paste moments but its limitation is that it says nothing about what happened in the chat before the paste. For example, two writers can both overlap with the AI exactly, but one might be pasting after asking for a single draft, and another might have iterated with the AI on tone and length for ten minutes and then pasted the final version. The score in this case would look identical. So the judges differences in their selections are not coming from whether or not a writer used or AI or not, but other cues that changed the way they chose the "human" document.

Section 2 covers the experimental design and the decisions made for each step in the experiment. Section 3 covers the platform, testing methods, and the recruitment methods I used to land participants. Section 4 presents Phase 1 results from the writers (agnostic of the judges evaluations). Section 5 covers the Phase 2 judge evaluation. Section 6 discusses the main takeaways from this study.

2 Experimental Design

Every design choice in this experiment was made for a reason. This section walks through each one and explains the thought process behind each decision layer.

2.1 Study Structure

Each participant sees only one version of the study. They are placed in either a treatment group (which sees an AI detection label) or a control group, and they complete two writing tasks: one arguing in favor of remote work and one arguing against it.

Condition assignment follows an alternating rule based on participant ID. Odd numbered IDs go to control; even numbered IDs go to test. This is deterministic rather than random, which is acceptable for a small sample where true randomization offers limited additional benefit. I wanted to make sure sample sizes were roughly equivalent across groups. Unit tests in the codebase confirm that the assignment logic produces exact 50/50 splits on condition for any sample size.

I also rotated the writing order so that half the participants argued in favor first and half argued against first. This is a standard technique called counterbalancing, where you spread out order effects so they do not pile up in one group. If every participant wrote their preferred stance first, I could not separate “wrote better because they believe it” from “wrote better because it was the first task and they were fresher.” The rotation follows a four-participant cycle that repeats for the full sample (Brooks, 2012). Table 1 shows how this works in practice. The drawback is that it does make the sample slightly noisier to read.

Table 1: Assignment logic. The four-participant cycle repeats indefinitely for equal session counts across conditions and writing orders.

PID	Condition	Task 1	Task 2
1	Control	For	Against
2	Test	For	Against
3	Control	Against	For
4	Test	Against	For

One limitation worth noting: for simplicity, I did not implement persistent sessions. If a participant dropped out due to internet or browser issues, their session would unfortunately be incomplete, and indeed this did happen. This was a deliberate trade-off to keep the platform simple for a small-scale study.

2.2 Treatment Design: The Detection Warning

The only difference between the two groups is a single informational notice in the briefing and on the submission page for each task. The test group sees the following text, displayed as a yellow banner on the task screen:

“Your final submission will be analyzed by an AI detection tool. Your AI detection score will be recorded and included alongside your submission in our research dataset.”

Task 1 of 2 — Briefing

In this task, you will write a **200–400 word opinion piece** arguing **IN FAVOR OF remote work**.

You will have **15 minutes** to complete this task.

How this works:

- **Left panel:** An AI writing assistant is available. You can use it for brainstorming, drafts, edits, or feedback.
- **Middle panel:** Your notepad. Use it to draft and refine your piece.
- **Right panel:** Submission. Paste your final version here and submit.

The timer will start when you click **I'm Ready** below.

Your final submission will be analyzed by an **AI detection tool**. Your **AI detection score** will be recorded and included alongside your submission in our **research dataset**.

Figure 1: The briefing for the test group. Notice the yellow tinted label for AI detection. Control *does not* see this yellow label.

This warning was designed carefully. My overall goal was to reduce as much behavioral and psychological bias as I could for the participants. Research on “priming” distinguishes between warnings that tell people what to do (behavioral primes, e.g., “there is this feature in the experiment and you should do this with it”) and warnings that activate a concept without prescribing a response (conceptual primes, e.g., “there is an AI detector”). Conceptual primes can still change behavior even when the prime itself does not specify an action (Bargh et al., 1996), and a meta-analysis of incidentally presented word primes confirmed that such effects are small but robust (Weingarten et al., 2016). My warning falls into that second category: it tells participants that detection will happen, but it does not tell them what “AI-sounding text” looks like or how to avoid it. The experiment does not even suggest there is any consequence to being detected, nor that this is part of any real evaluation. That ambiguity is the point.

I also had to be careful not to make the warning too aggressive. Research on reactance shows that overly forceful warnings can backfire, causing people to do the opposite of what you intended (Brehm, 1966). For this experiment, I did not choose a large red banner with alarm language because that could have pushed participants to use the AI with less or more extreme intent (even going as far as defying the experiment), or made them so anxious that their behavior would no longer reflect anything natural. I chose a yellow background with neutral phrasing to keep the warning noticeable

but not alarming.

The label sits in 2 positions: one within the first briefing just above the forced chatbot interaction to onboard the participant (preventing cases where a participant might miss the label, Figure 1), and the second moment is placed just above the “I am ready to submit” checkpoint to let the user see it again before submitting (Figure 3).

When designing the experiment, I considered how the user might interact with the treatment itself when it comes to intent. The treatment mirrors the kinds of warnings students might encounter in real academic settings rather than creating an artificial challenge to beat the detector. If the test group was motivated to defeat an AI detector, it would lead to high-intent changes in behavior. I want to observe how people naturally responded to the idea of being monitored, rather than feeling like they are playing a game.

2.3 Demand Characteristics and the “Observer Effect”

In any experiment, there is a risk that participants figure out what you are testing and change their behavior to match what they think you want. This is called “demand characteristics” (Orne, 1962). There is also another framework that separates participants being aware of the hypothesis, being motivated to comply, and having a strategy for responding. Awareness alone does not always lead to behavioral change (Corneille & Lush, 2023). A larger-scale test with over 12,000 participants found that explicitly telling people the hypothesis did not *actually* alter treatment effects, which suggests demand effects may be less threatening than commonly assumed for this type of experiment (Mummolo & Peterson, 2019).

Still, I built in several protections to keep the test valid:

1. **Between-subjects design:** Each participant sees only one version of the experiment, so they have nothing to compare against.
2. **Cover story:** The study is framed as research on “how people use AI writing tools” when recruiting volunteers (which is true but broader than the actual hypothesis).
3. **Unobtrusive primary outcome:** The main thing I analyze is the submitted text itself, which participants produce naturally as part of the task.
4. **Informational warning:** The treatment does not tell participants what the “correct” response is.
5. **Post-experiment hypothesis probe:** After finishing, I ask “What do you think this study was about?” This lets me check whether anyone guessed the real purpose and filter accordingly. (No one has.)

There is also a baseline observer effect to consider. Both groups know they are in a study, which can change behavior on its own. This is a common issue, however, a systematic review of 19 studies found that there is no single observer effect (McCambridge et al., 2014). In this case, both groups experienced the same baseline awareness of being observed. The test group received a second layer of observation awareness (the AI detection notice). So I am measuring the *incremental* effect of the detection warning above whatever baseline shift comes from being in a study at all.

The closest parallel I found to this “AI Detector” label in practice is plagiarism detection software like Turnitin. A study of medical students found that introducing plagiarism detection software alongside penalties reduced plagiarism rates from 21% to 2% across successive cohorts (Bilic-Zulle et al., 2008). A similar study at the University of Botswana found that warning students about Turnitin reduced plagiarism from 20.5% to 16.2%, with the biggest drops among the most egregious

cases (Batane, 2010). But there is an important difference between this study and the plagiarism research: with plagiarism, students know exactly what to avoid (do not copy text). In this study, participants are ambiguous about what the detector targets. They do not know what “AI-sounding text” looks like. That means the warning creates awareness without giving participants a clear strategy for responding, which is what makes the behavioral response interesting to study. The limitation of this study however is that it is inconsequential to the writers. There is no real penalty for using AI here, unlike a school or a graded requirement.

2.4 Writing Task Parameters

Each participant writes two opinion pieces on remote work, one arguing in favor and one arguing against, with a target length of 200 to 400 words per piece.

Topic choice. I chose remote work because it requires no specialized knowledge, most people have an opinion on it, and it is polarizing enough that writing against your own belief takes real effort. That cognitive demand is intentional: it is exactly the situation where reaching for the AI assistant becomes most tempting.

Word count. The 200 to 400 word target was intentionally set based on a few pieces of evidence backed in prior research. Adults composing original text produce roughly 19 words per minute (Karat et al., 1999). With planning and pausing over 15 minutes, that works out to roughly 200 to 350 words unassisted. Research on AI-assisted writing found that AI tools reduce task completion time by about 40% (Noy & Zhang, 2023), which in a fixed time window means roughly 280 to 490 words. On the analysis side, lexical diversity indices become more stable as text length increases, with MTLD performing reliably on texts as short as 50 words (Zenker & Kyle, 2021), so the 200-word floor comfortably supports the NLP features I planned to extract without overwhelming participants. A 400-word ceiling keeps the task achievable in 15 minutes but prevents participants from just dumping raw AI output. The constraint forces them to at least consider editorial choices about what to keep and what to cut, and that curation is itself a signal worth measuring.

Timer. A soft 15-minute countdown runs on screen for each task. It is not enforced: participants can continue writing after it reaches zero. Participants are not aware of this soft countdown. The timer creates psychological time pressure without risking data loss from a hard cutoff. Actual duration is logged as a variable for analysis.

The other challenge is encouraging the use of the AI agent. Forcing behavior with an AI agent is tedious to implement, and it would also bias results by forcing individuals to write with an AI. Instead, the briefing prior to the actual task forces a small demo / tutorial to encourage the use of the AI assistant.

2.5 Survey Design

Pre-survey. Before writing, participants also report their name or alias (anonymity is allowed), their personal stance on remote work, their AI usage behaviors, and their education level.

The stance question uses a four-point scale with no neutral option: Slightly Favor, Favor, Slightly Oppose, Oppose. The neutral option is omitted on purpose. Research on survey design shows that including a “no opinion” or “neutral” midpoint encourages “satisficing,” where respondents pick the middle option to avoid thinking about it (Krosnick et al., 2002). Removing it forces everyone to lean one way, which gives a clean way to classify each document as belief-aligned (the assigned

stance matches their real opinion) or belief-misaligned (it does not). That classification is necessary for the belief-alignment analysis.

The AI usage question is multi-choice rather than a single frequency measure. Participants check all that apply from a list that includes “search engine use,” “writing help,” “coding and technical tasks,” “work or school assignments,” “tried but irregular use,” and “does not use AI tools.” This captures *how* someone uses AI, not just how often. A participant who uses AI for coding but never for writing will likely interact with the writing assistant differently than someone who uses AI for writing every day. For this sample size that part is practically negligible, but I kept it for completeness.

Post-survey. After writing both tasks, participants self-reported how much they relied on the AI for each piece using a five-point ordinal scale ranging from “I wrote it entirely myself” to “AI wrote almost all of it”

The ordering of post-survey items mattered. Research on question order effects shows that structured scales can frame how people respond to open-ended questions if asked first (Schwarz, 1999). I put open-ended questions before structured ones, and the hypothesis probe (“What do you think this study was about?”) always comes last so it cannot contaminate anything else.

3 Platform, Implementation, and Data Collection

3.1 Platform Architecture

The experiment runs as a Streamlit app backed by the Anthropic API (Sonnet was used as the base model) for the chatbot. All participants access the same URL and enter a shared access code. The application assigns conditions automatically based on the participant landing sequence, manages the session flow, and writes data to a Supabase PostgreSQL database in real time so I could analyze the results and eventually propagate them into Phase 2 for the judges.

The session goes through nine stages in this static order:

1. Welcome and consent
2. Pre-survey (demographics and stance)
3. Briefing for Task 1 (test group sees the detection warning here)
4. Task 1: writing with AI assistant
5. Transition screen
6. Briefing for Task 2
7. Task 2: writing with AI assistant
8. Post-survey (self-reported AI reliance)
9. Thank-you screen

The task screen uses a three-column layout. The left column contains the AI chat interface, where participants can send messages to Claude and receive responses in real time. The center column is a rich-text notepad for drafting. The right column holds the submission text area with a live word count and a submit button. This separation between drafting space and submission space forces participants to make a deliberate editorial choice about what to submit rather than pasting raw AI output directly.

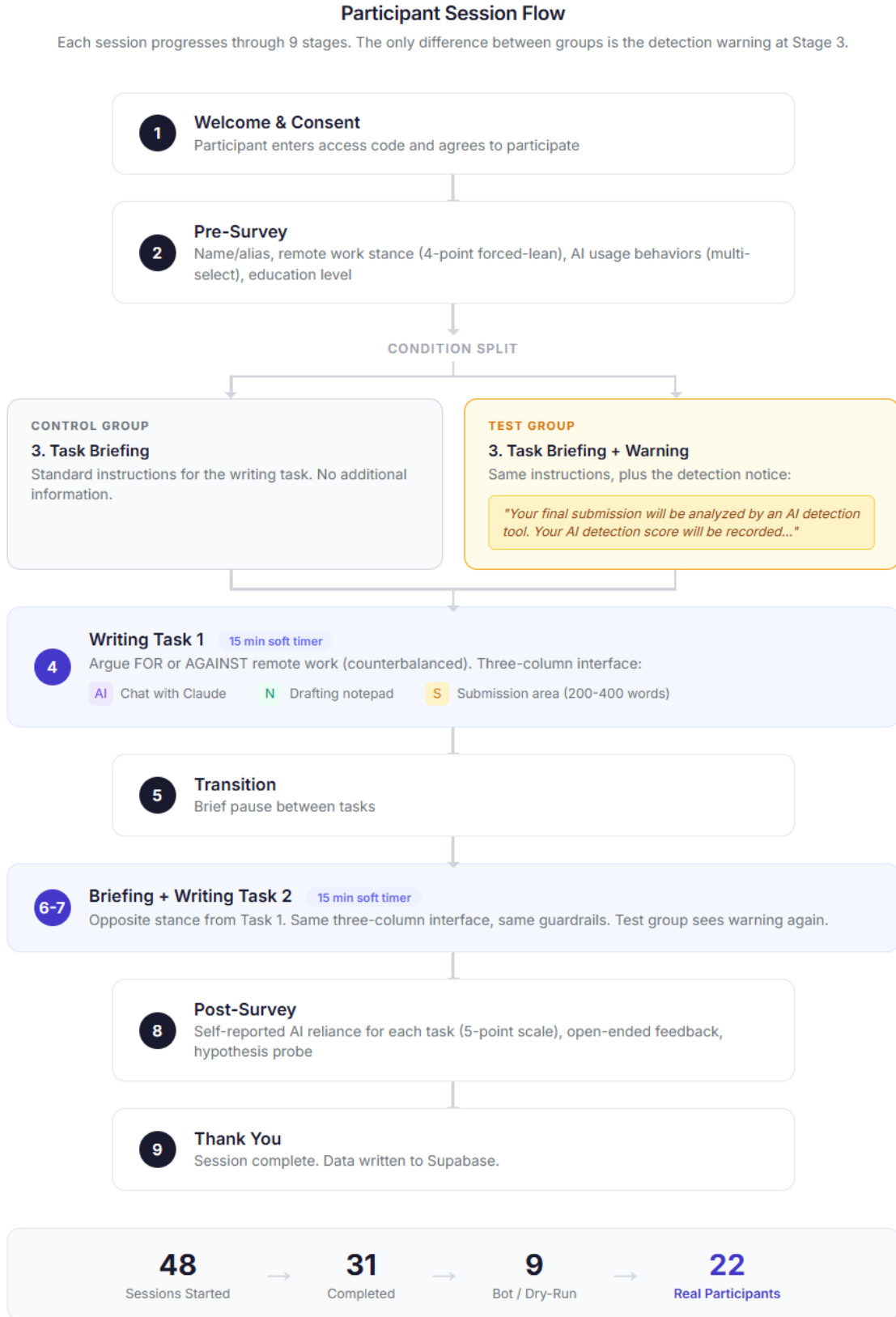


Figure 2: Participant session flow. Each session progresses through nine stages. The only difference between groups is the AI detection warning shown at Stage 3.

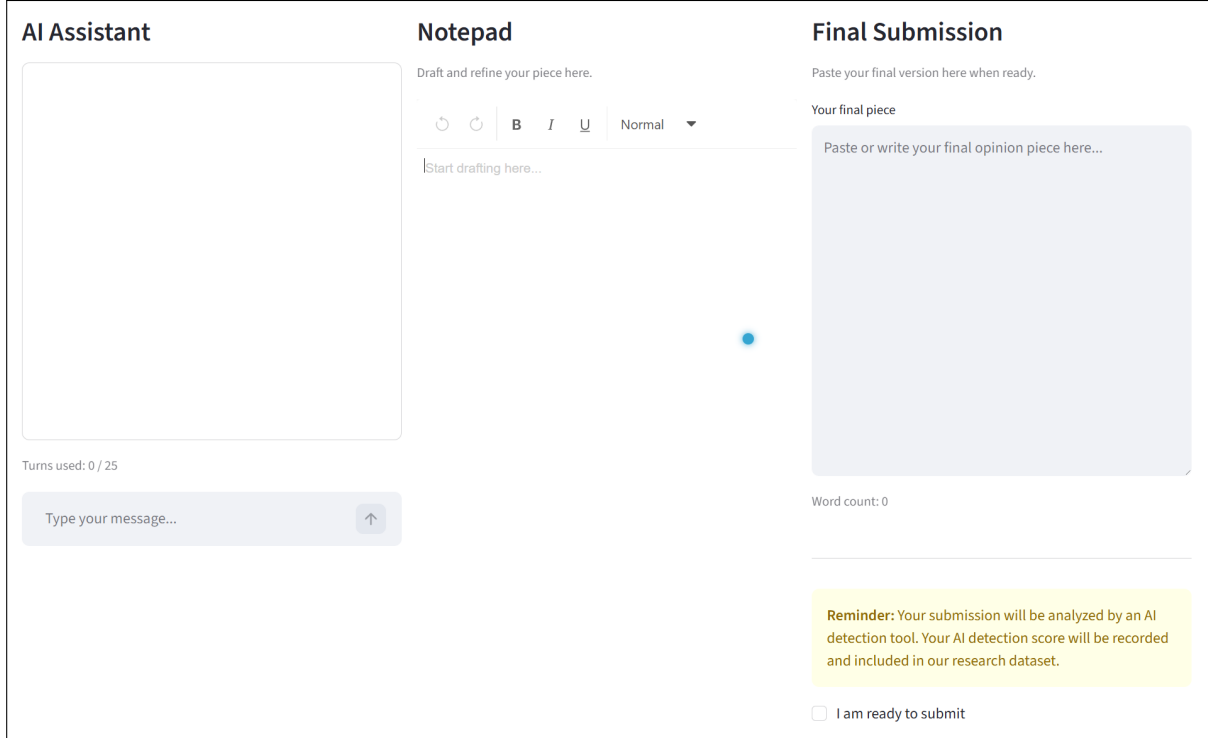


Figure 3: Both groups see the same three-panel setup, with an AI chatbot, a notepad for edits, and the final submission screen. Users confirm with the button if they are ready to submit to avoid accidents. Only the test group sees the AI detection label.

3.2 Recruitment and Data Collection

Participants were recruited through personal outreach via social media (Instagram, LinkedIn) and direct messaging (WhatsApp), targeting friends, classmates, and broader social networks. The response rate among even close acquaintances was modest—a reminder that the goodwill of friends, while freely given in conversation, becomes a scarcer resource when a hyperlink and twenty minutes are involved.

Data is stored in Supabase, a hosted PostgreSQL service. Each session is a single row containing the full interaction record: pre-survey responses, conversation logs with timestamps and token counts for every message, submission text and word count, post-survey responses, and session-level metadata (start time, completion time, total tokens used).

3.3 Completion Funnel

Phase 1. Of the sessions initiated during the collection period, a subset reached the “thank-you” screen. After filtering out bot dry-runs and incomplete sessions, the final analytic sample is **21 real Phase 1 participants** producing **41 validated documents** (one participant had a single completed task, the rest contributed both). The dropped sessions were participants who entered the access code but did not finish, typically due to time, loss of interest, or mobile-browser issues.

Phase 2. For the judge evaluation, **315 judge sessions were started, 76 sessions completed fully**, and together they produced roughly **2,000 pair-level responses** (partial completions included, since each individual pair response is still usable data)

4 Phase 1 Results (RQ1)

This section covers the EDA of the Phase 1 participants and the 41 validated submissions. Its purpose is to answer research question 1. Do the two groups within the study produce different writing patterns and writing outputs? With 9 control and 12 treatment participants completing the study, the sample is underpowered to run any statistical tests on the differences between both groups. However, there are still basic descriptive patterns that show both groups treated the study differently.

4.1 Sample Demographics

The sample is skewed toward writers with a graduate-level education, reflecting the Georgia Tech and personal-network recruitment pool from social media and messaging forums. The "Remote-work" stance is heavily favorable: most completed participants selected "Favor" or "Slightly Favor," with only a small minority selecting the opposing side. The practical consequence is that most "Against" documents were written in a belief-misaligned condition, which the judges might mark differently. Condition assignment produced 9 control and 12 treatment participants (18 and 23 validated documents respectively since participants complete two documents). The imbalance is the result of session drop-off or other incomplete submissions being removed, not of the assignment rule itself. Section 2.1 describes how a deterministic rule in the Streamlit app would ideally produce an exact 50/50 split on any complete sample.

4.2 Chatbot Interaction by Condition

Table 2 summarizes the main chatbot-interaction metrics per task, split by condition, and Figure 4 shows the same data as box plots.

Table 2: Chatbot interaction metrics by condition. Turns, duration, and word count are per task; session total tokens are per participant.

Metric	Control ($n = 9$)		Treatment ($n = 12$)	
	Mean	Median	Mean	Median
Conversation turns (per task)	2.4	2.0	3.0	2.5
Task duration (seconds)	571	560	693	668
Submission word count	275	260	278	288
Session total tokens	5,411	5,010	8,717	4,876

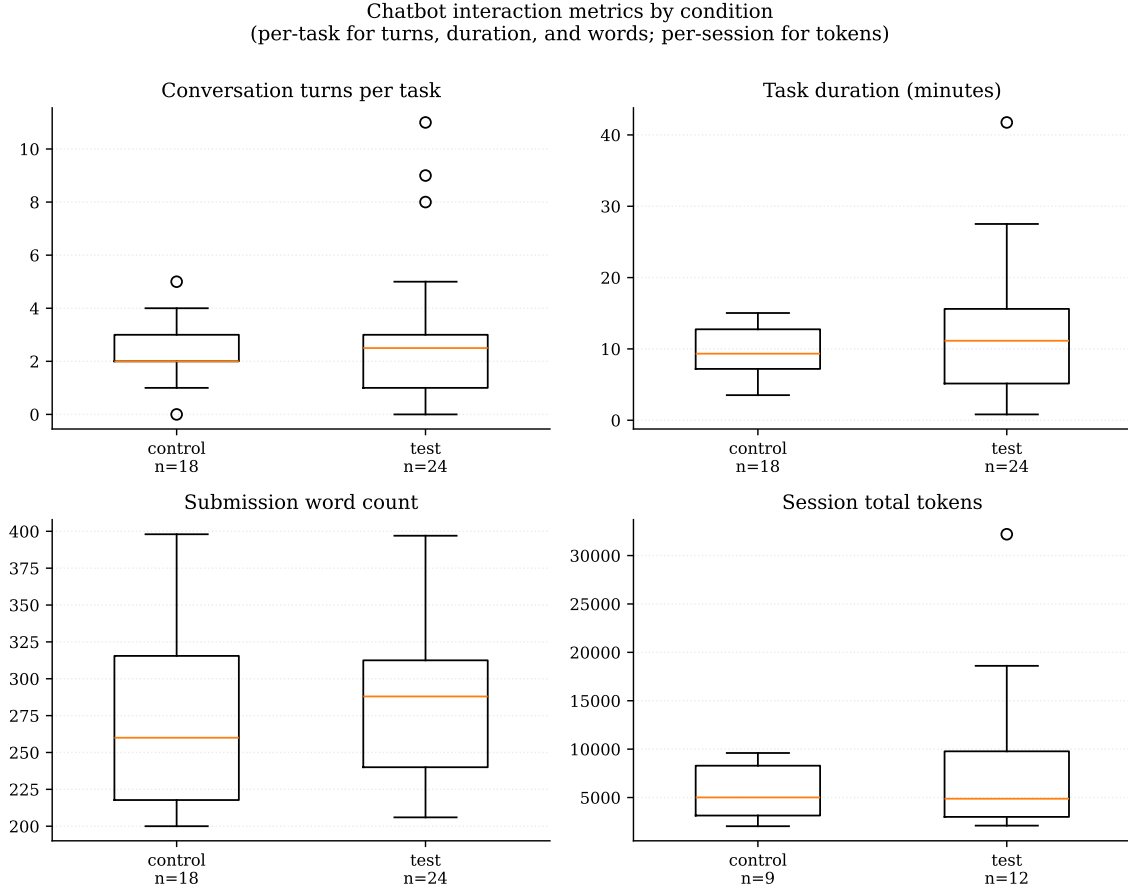


Figure 4: Chatbot interaction metrics by condition. Turns, duration, and submission word count are reported per task (two tasks per participant). Session total tokens are reported per participant. Per-task counts reflect all completed session tasks before document-level validation filtering (one test-group task was excluded during validation, reducing the validated document count to 23).

Three descriptive patterns are visible. First, treatment participants take more conversation turns per task than control participants on both the mean and the median. Second, treatment participants generally spend longer on each task by about two minutes at the median. Third, session total tokens are higher on average for the treatment group but with much wider variance, driven by a few participants who had long back-and-forth sessions with the chatbot. Word counts are nearly identical across both groups, which is expected given that the task prompt bounded the length target at 200 to 400 words. Qualitatively, some users simply prompted the chatbot to give it an essay spanning the word count. These patterns answer RQ1 at the descriptive level: the two groups do interact with the chatbot differently, with the treatment group using it more intensively.

4.3 AI Overlap with Chatbot Output

To quantify how much of each submission came from the chatbot, I feature engineered an AI overlap score comparing the submitted text to the assistant messages in that participant’s conversation log.

Motivation and design rationale. The simplest approach to “how much of the submission came from the chatbot?” is exact word-for-word matching. I avoided using this metric alone because

this doesn't capture edge cases in two distinct ways. First, a writer can keep the AI's structure and substitute synonyms or different phrases. This won't be captured. A writer can also try reordering text, and this only counts the sentences that survived in order. A "score" for this setting therefore needs to combine multiple similarity metrics rather than rely on one. Document fingerprinting techniques use n-gram hashing to detect reused passages across texts (Schleimer et al., 2003), and large-language-model training-data decontamination uses similar n-gram overlap and exact substring matching via suffix arrays (Brown et al., 2020; Lee et al., 2022). I adapted the general principle of combining multiple text-similarity measures to engineer a robust scoring function.

Definition. Let's make S a participant's submission text, (lowercased and with whitespace normalized), and let $A_1, A_2, \dots, A_k$ be the assistant messages (or turns) in that participant's conversation log for the same task, preprocessed the same way. For any single assistant message " A ", I compute three similarity scores between S and A . Each one targets a different way a writer might reuse AI text. This is to capture the AI overlap the user submitted.

The first score looks at short phrases. We can use a type of sliding-window method to capture similarity. For example, Let $T_3(x)$ be the set of three-word windows in x (for example, the sentence "the dog sat down" contains the windows "the dog sat" and "dog sat down"). I measure what fraction of the submission's three-word windows also appear somewhere within the AI messages sent to the user. This is very similar to a Jaccard Score but using full-words instead of n-grams:

$$t(S, A) = \frac{|T_3(S) \cap T_3(A)|}{|T_3(S)|}. \quad (1)$$

A high value here flags short phrase-level reuse, even when the writer placed fragments from different AI responses together.

The second score looks for one long paste. I find the longest run of characters that appears identically in both S and A , call its length $L(S, A)$, and divide by the length of the submission. This is identical to looking at a word-for-word overlap assuming they just copy-pasted and didn't make any edits:

$$\ell(S, A) = \frac{L(S, A)}{|S|}. \quad (2)$$

A value near 1 means almost the entire submission is a single copy of AI text. The same exact function is used in plagiarism detection (Schleimer et al., 2003) and in large-language-model training-data decontamination (Brown et al., 2020; Lee et al., 2022).

The third score captures total overlap, including pieces that have been shuffled around. Using a standard sequence-matching routine from the Python standard library, I find every non-overlapping run of characters that appears in both S and A and sum their lengths, giving $M(S, A)$.

$$r(S, A) = \frac{2M(S, A)}{|S| + |A|}. \quad (3)$$

This score would stay closer to 1 when a writer kept most of the AI's sentences or words but rearranged sentences or paragraphs. The previous two measures would each miss on their own.

To summarize how much S resembles a given AI message A_i , I take the largest of the three scores:

$$AI \text{ overlap Score} = \max(t(S, A_i), \ell(S, A_i), r(S, A_i)). \quad (4)$$

The AI overlap score for the document is then the largest of the set of scores that we calculate. A score near 1 means the submission is exactly a paste of one of the AI responses the writers had with the chat. A score near 0 means the submission shares no meaningful text with anything the AI produced, or the participant never actually used the chatbot.

Why three measures and not one? Each of the three checks a different way a writer might reuse AI text: small phrases, long chunks, or the full AI output reordered in some way. Taking the maximum is the conservative choice. The idea is that a writer cannot circumvent the scoring if they utilized the AI by rearranging or lightly paraphrasing. However, there may be some edge cases or limitations to these methods, but I used them to cover the main use-cases of using AI-assistance for writing.

What the score shows. Figure 5 shows the overlap-score distribution by condition. Figure 6 shows the pooled histogram across all 41 documents.

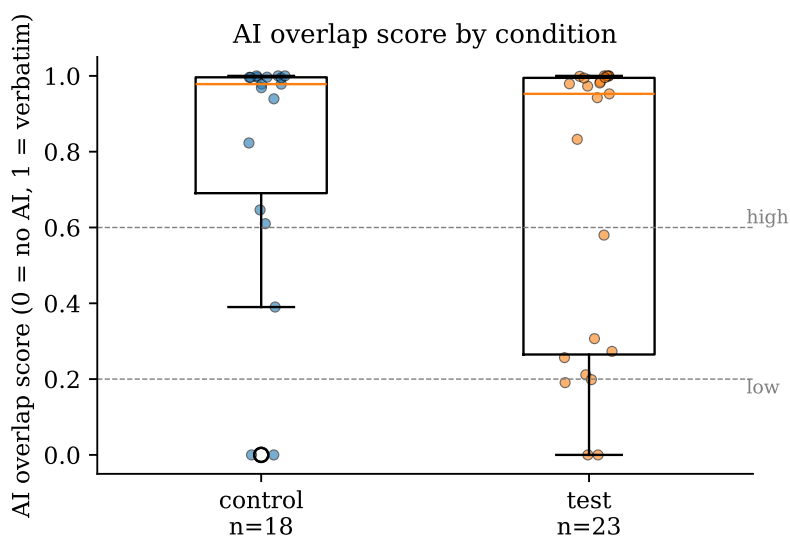


Figure 5: AI overlap score by condition. The score is the maximum similarity between the submission and any single AI response across three measures. Note how the test group has more cases where users used AI infrequently or not at all for their writing

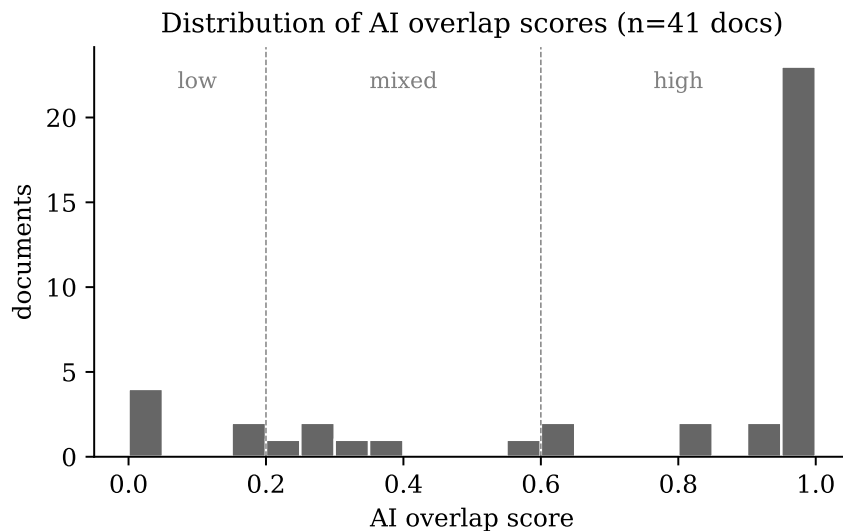


Figure 6: Distribution of AI overlap scores across all 41 validated documents. The distribution for the writers is concentrated on many writers who used AI verbatim in some way according to the three measures, while some documents used more editing and a handful used none.

Only six documents sit at or near zero. Of these, three participants never sent a user message to the chatbot on one of their tasks, one participant sent a message but the session recorded no assistant response (this was a streamlit bug), and two participants iterated with the chatbot but retained little of its output in the submission. The remaining 35 documents concentrate above 0.6, with a mode near 1.0. The control and treatment medians are 0.978 and 0.953 respectively, nearly identical. At the descriptive level, most writers in both conditions pasted at least one AI response close to verbatim, and a minority in each condition either did not engage the chatbot at all or rewrote substantially.

4.4 Stylistic (Stylometric) Features

Table 3 lists the stylometric features extracted from each submission. Each feature has an established role in stylometry or AI-text detection and is cheap to compute. Figures 7 and 8 show four of them as box plots by condition.

Table 3: Stylometric features computed per submission. N is the number of words and N_s the number of sentences. “per 1k” rates multiply the token count by $1000/N$.

Feature	Definition	What it captures / source
Type-Token Ratio	$TTR = V /N$, where V is the set of unique tokens	Lexical diversity. AI output tends to reuse the same words more often within a response (Reviriego et al., 2024).
Mean sentence length	$\bar{L} = N/N_s$	Baseline rhythm. AI output tends toward longer, more uniformly structured sentences.
Sentence-length SD	σ_L across sentences	Variation in rhythm. Human writers mix short and long sentences more often; AI output tends to be more uniform (Desaire et al., 2023).
First-person rate (per 1k)	$\text{count}(I, me, my, we, our, \dots) \cdot 1000/N$	Personal voice. AI output typically uses fewer first-person pronouns than human writing (Sandler et al., 2024).
Hedging rate (per 1k)	$\text{count}(perhaps, might, possibly, \dots) \cdot 1000/N$	Tentative language. Epistemic stance, as reflected in lexical bundles, differs in frequency between human and AI text (Jiang & Hyland, 2025).
Contraction rate (per 1k)	$\text{count}(don't, can't, it's, won't, \dots) \cdot 1000/N^1$	Casual tone. AI output often avoids using contractions, preferring formal phrasing like “do not” instead of “don’t.”

¹The contraction regex also matches possessive forms (e.g., “student’s”). Both groups are affected equally, so this does not bias the comparison, but the rate slightly overstates true contraction frequency.

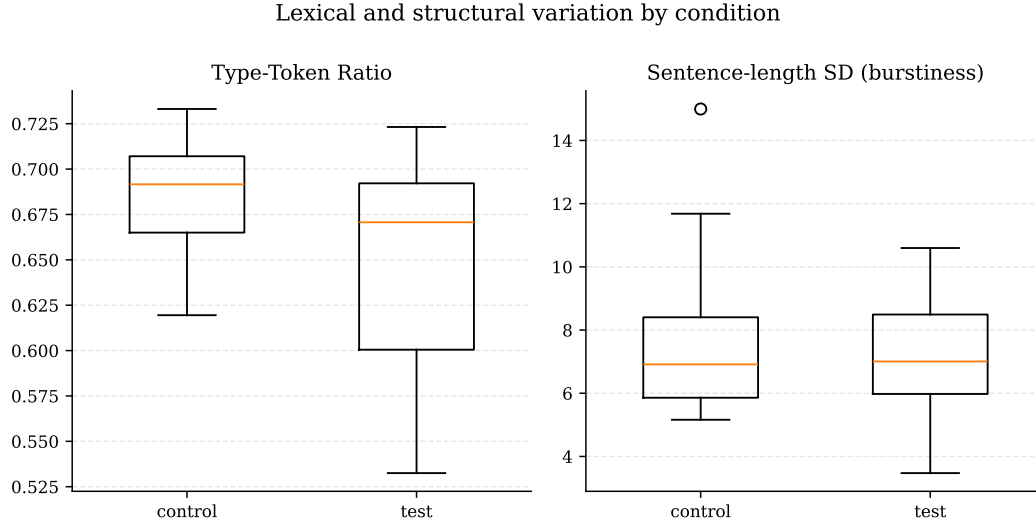


Figure 7: Lexical and structural variation by condition: type-token ratio and sentence-length SD (burstiness). Distributions overlap heavily between control and treatment.

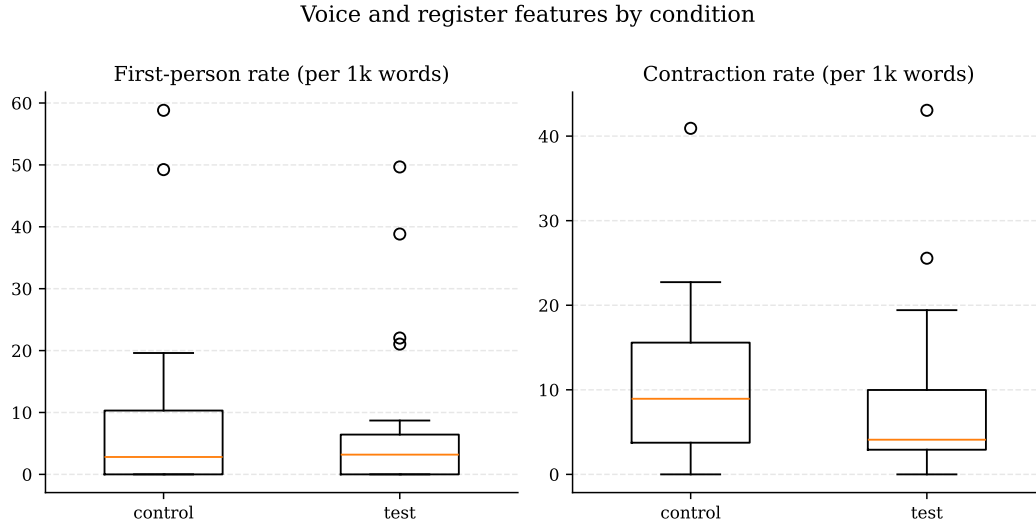


Figure 8: Voice and register features by condition: first-person rate and contraction rate (both per 1,000 words). Distributions overlap heavily between control and treatment.

4.5 Summary of Phase 1 EDA

My research question 1 asked whether writers behave differently under the threat of AI detection. With such a small sample size of writings, we can't run any statistical tests to genuinely prove the treatment group behaved differently in the study compared to the control group. However, based on the EDA and the analysis of different features between both groups, there are differences in behaviors between both. This is where evaluation from the judges for these documents becomes critical to the success of this experiment. The judges *become* the statistical test of this study - where they evaluate whether or not the behavior of the treatment group would create more "human-like" submissions.

The experiment design almost intends to compensate for the lack of power in a statistical test between both groups - and the judges become a secondary layer from a human perspective to determine whether a behavior difference occurred when the treatment group saw the AI-detection label. In phase 1, I found that the treatment group takes more conversation turns, spends more time on each task, and exchanges more tokens per session than the control group. But the submissions completed tell us that their usage of AI text is almost identical. Control and treatment AI similarity scores differ by only 0.025 between the medians, and the stylometric feature distributions overlap as well. Both groups relied heavily on the AI chatbot to complete the writing within the word-count and the time allotted, but the treatment group was shown to be more variable in their interactions with the AI - some completely abandoning the chatbot, or some iterating extensively to deliver a submission. Phase 2 becomes the statistical test for the difference between these two groups. Where judges will compare both documents under a deceptive design and asked "Which document was written by a human" - despite both being written by AI. The hypothesis is that the group seeing the AI-detection threat will be interpreted as more likely to be "human" written compared to the control group's submissions.

5 Phase 2: Human Judge Evaluation (RQ2)

5.1 Recruitment and Completion

I recruited judges through four channels over a three-week period: Reddit ads (32,000 impressions, 1,200 post landings, 324 survey clicks), LinkedIn posts across four rounds (6,400 total impressions), direct WhatsApp messages to roughly 300 contacts, and Instagram posts (approximately 80 viewers). Attribution is untraced because the survey link was identical across all channels, but Reddit was the dominant source by volume. Figure 9 shows the recruitment funnel.

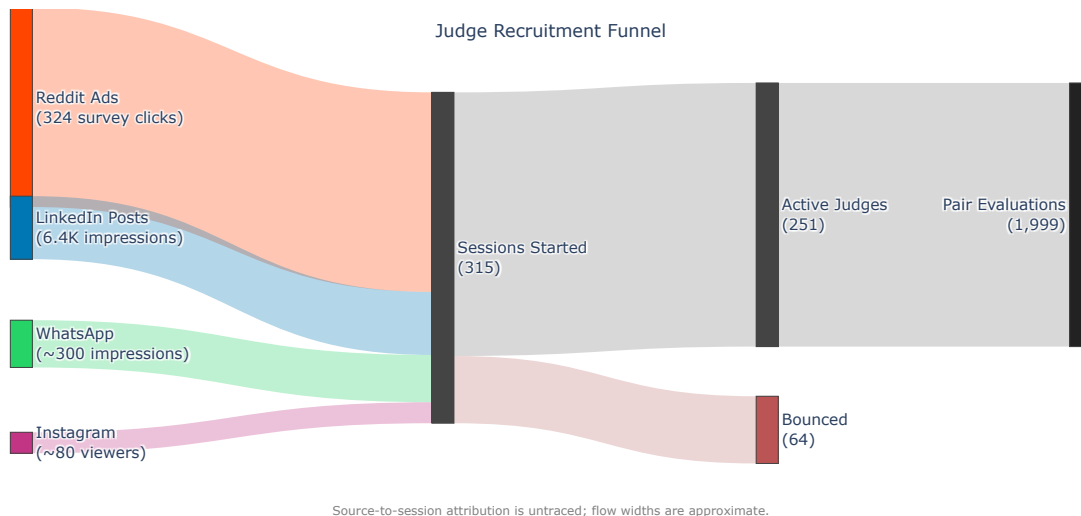


Figure 9: Judge recruitment funnel. Session attribution is untraced; flow widths from the four channels are approximate.

Of 315 sessions started, 251 judges submitted at least one evaluation, producing 1,999 total pair-level responses. Each pair showed one control document and one treatment document for the same stance (both "For" or both "Against" remote work), with the left/right position randomized. The

judges saw a 150-word preview of each document with a forced 10-second reveal period before they could choose, acting as a guard-rail against automatic responses and scripting. A judge could also optionally expand to read the full text. The survey asked one question per pair: “Which one was written by a human?” with an optional confidence rating.

5.2 Detection Accuracy

Across all 1,999 paired evaluations, judges selected the treatment (warned) document as the human-written one 54.13% of the time, versus 45.87% for the control (unwarned) document. A two-sided binomial test against the chance rate of 50%, treating each response as independent, rejects the null hypothesis that this is a random guess at $p = 0.000243$ ($\alpha = 0.05$). The 95% Wilson confidence interval on the control-as-human rate is [43.70%, 48.06%], which sits entirely below the 50% chance line. The effect seen is consistent across both stances. FOR-stance pairs show a 46.00% control-as-human rate ($p = 0.012$, $n = 1,000$), and AGAINST-stance pairs show 45.75% ($p = 0.008$, $n = 999$).

Figure 10 shows these intervals. All three (overall, FOR, AGAINST) fall to the left of the 50% chance line, showing that the effect is not driven by one stance and is statistically significant.

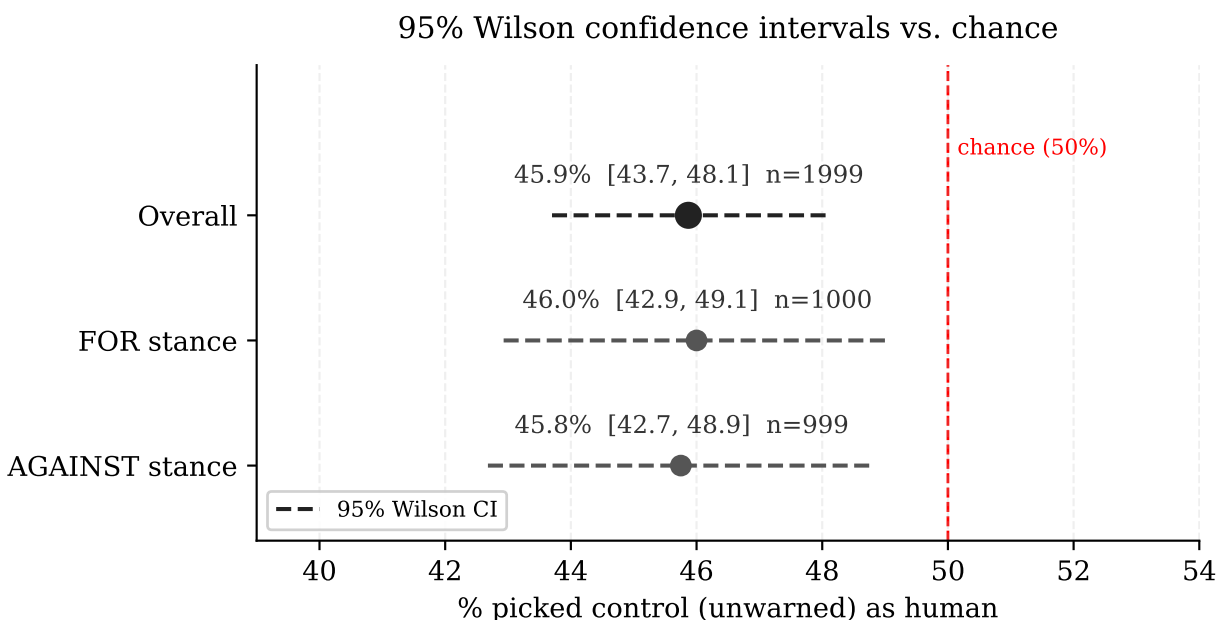


Figure 10: 95% Wilson confidence intervals for the rate at which judges picked the control (unwarned) document as human. The dashed red line marks the 50% chance rate. All intervals fall below 50%, consistent with the hypothesis that warned writers are perceived as more human.

The hypothesis was that writers in the treatment group who saw the AI-detection warning would edit their submissions more carefully to sound human, and that judges would therefore identify the warned (treatment) document as human at a higher rate. The observed effect is in exactly that direction. The magnitude is small, but it is statistically significant across both stances.

5.3 Confidence, Reading Behavior, and Fatigue

Confidence. After each pair, judges had the option to rate their confidence on a scale from 1 to 5 (1 being a guess, 5 being very confident). Higher-confidence judges were more likely to pick the treatment document as human. At confidence levels 4 and 5, the control-as-human rate drops to 42 to 43%, compared to the overall 45.87%. Figure 11 shows the breakdown by confidence level.

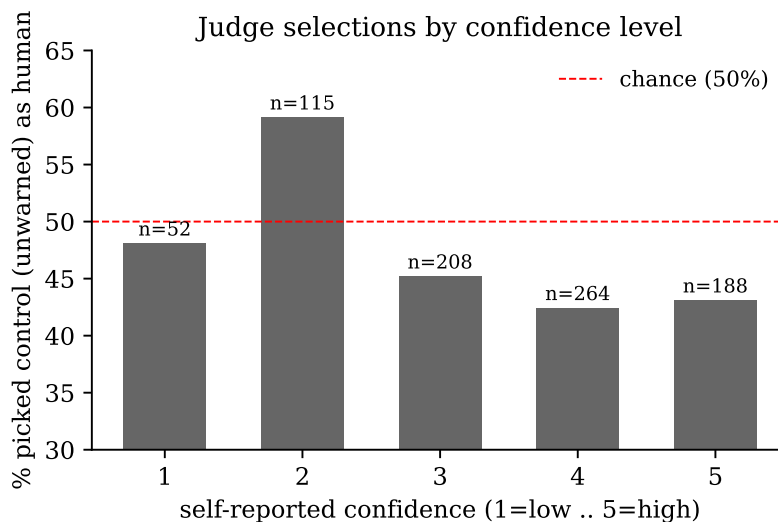


Figure 11: Control-as-human rate by judge self-reported confidence (1 = guess, 5 = confident). Higher-confidence judges leaned more toward selecting the treatment document as human.

Reading time. Judges who spent more time on a pair were more likely to pick the treatment document as human. The fastest quartile (under 10.6 seconds) was the closest to being near chance at 51.6%, which is intuitive when judges were only briefly glancing at the writings rather than reading thoroughly. The three slower quartiles ranged from 43% to 45%. Figure 12 shows the breakdown by time quartile.

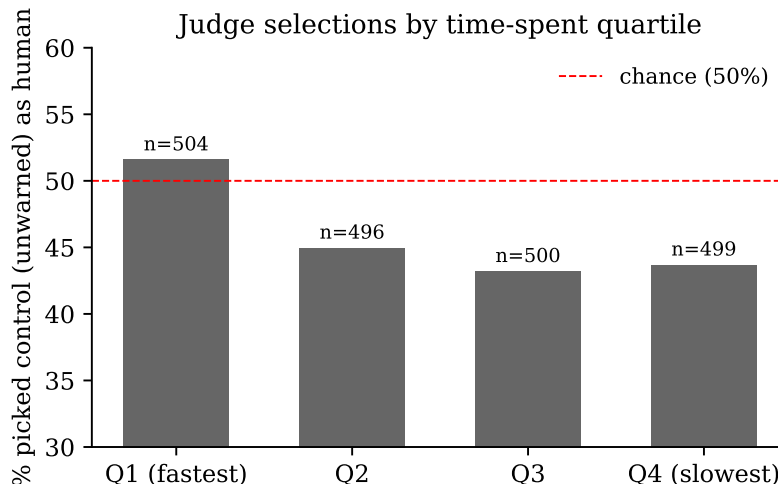


Figure 12: Control-as-human rate by time-spent quartile. The fastest responders (Q1, under 10.6s) are near chance; slower readers lean toward selecting the treatment document as human.

Document expansion. Most responses (76.3%) involved no expansion of the document at all. This meant judges decided from the 150-word preview alone in the majority of cases. Among the 92 responses where the judge expanded *just* one of the two documents, the control-as-human rate dropped to 35.9%. Meaning that when a judge expanded a writing to investigate, they were able to pick the treatment document as human 64.1% of the time. What is interesting is that when judges expanded both documents - this brought the rate back to 45.8%. This suggests that selectively reading one document in full may amplify whatever cue judges are picking up on. However, given the small sample of judges who deliberately expanded documents, it is possible that this finding is simply noise - a judge may have expanded documents just to check if the UI worked rather than actually reading the expanded text.

Fatigue over time I split each judge’s responses into terciles (early, middle, late) to check whether detection accuracy changed over the course of a session. The control-as-human rate was 45.87% for early pairs, 45.74% for middle, and 46.03% for late. There is no clear fatigue effect present. Judges didn’t get better or worse over time, and the signal held steady throughout each session.²

5.4 Per-Document Analysis: Is the Effect Driven by Outliers?

I checked the concern that it was possible within a small document pool that one or two outlier documents are driving entire experimental result. To check, I computed the rate at which each of the 41 documents were chosen as human across all the pairs it appeared in.

²This analysis was motivated by qualitative feedback from a Reddit commenter during judge recruitment who suggested that detection ability might degrade as judges reviewed more pairs.

Table 4: Per-document chosen-as-human rates, summarized by condition. The effect is distributed: most treatment documents sit above 50% and most control documents sit below it.

Condition	Docs	Mean rate	Docs above 50%	Range
Control	18	45.9%	5 of 18 (28%)	29.8% to 62.4%
Treatment	23	54.0%	15 of 23 (65%)	30.5% to 76.5%

The strongest outlier is participant P26 (treatment), whose two documents were chosen as human 72% and 76% of the time. However, this makes sense entirely. P26 *never* sent a single message to the AI chatbot. They wrote both submissions entirely on their own, and their text reads accordingly. They have frequent contractions, personal phrasing, and variable sentence lengths. Judges may have recognized this as distinctly human. Removing P26 from the dataset weakens the overall effect (control-as-human rate moves from 45.87% to 47.89%) but *does not* eliminate it, and the broader pattern holds: 15 of 23 treatment documents were chosen as human above the 50% chance line, compared to only 5 of 18 control documents. The effect is distributed across the document pool, not concentrated in a single outlier.

5.5 Summary of Phase 2

Phase 2 confirmed the original hypothesis: judges identified the warned writer’s document as human at a rate above chance. The effect is small but statistically significant and holds across both stances, across confidence levels, and across the full session without fatigue. No single document drives the result.

6 Discussion and Conclusion

We reject the null hypothesis that there is no difference between the treatment (warning) group and the control (no warning) group. Judges identified the treatment document as "human" at a statistically significant rate more often than they identified the control document as being human. This is consistent with the original hypothesis for this experiment. The effect is small but significant and holds across both stances - both FOR and AGAINST.

What is interesting to note is given a warning of AI Detection, participants *did not* necessarily shy away from using an AI chatbot to assist them in their writing. Instead, the *opposite effect* was seen. Writers spent more time, took more turns, and used more tokens than the control group. Several treatment participants abandoned using the chatbot entirely, choosing to write the whole submission themselves. No control group participant did this. The warning *didn't* mitigate the use of AI; it actually polarized writers to lean harder into using AI or opting out completely.

Yet, for all that behavioral difference between the two groups, the submissions themselves were nearly identical when measured for AI similarity to what was used in the chats. AI scores only differed by 0.025 between medians. Every stylistic feature showed significant overlap between both groups. The judges however chose the treatment document as "human" more often, at a statistically significant rate across 2000 paired evaluations.

The implication here tells us that warned writers difference in behavior - using AI or not - changed the way their submissions were interpreted by human judges. This could mean a stylistic rhythm, or a personal opening, or a different prose within their conversations with the AI chatbot or their own

drafts, is not a measurable feature analytically - but a human judge could identify the difference themselves.

Feature-based AI detection tools may extract typical AI signals, but this experiment suggests that humans given an AI-detection flag can actually lead to submissions that can convince judges to choose these submissions as more 'human' sounding.

6.1 Limitations

Several limitations constrain generalizability. The sample of 21 writers is small, skewed toward graduate students, and recruited from personal networks. The 9-vs-12 condition split limits statistical power for within-Phase 1 comparisons. The writing task (opinion pieces on remote work) may not generalize to other genres or domains. The AI overlap score captures surface-level textual reuse but not deeper forms of AI reliance such as idea generation or structural borrowing. The warning carries no real consequence, unlike academic integrity policies; a consequential warning might produce stronger effects. The judge task is inherently noisy—while the aggregate effect is significant, individual judge accuracy is only slightly above chance.

We assume each observation and judgment of the sample pairs is independent from another, but responses are clustered by both judge (251 judges contributing ~ 8 responses each) and document (41 documents each appearing in many pairs). The effective sample size is therefore smaller than the 1,999 responses reported, and the binomial p-value ($p = 0.000243$) may be anti-conservative. A mixed-effects logistic regression with crossed random intercepts for judge and document would provide a more conservative test. We note, however, that the effect is consistent across multiple independent decompositions—by stance, by confidence level, by time quartile, and per-document—which provides converging evidence beyond the single omnibus test. A formal mixed-effects analysis is planned for future work.

References

- [1] Bargh, J. A., Chen, M., & Burrows, L. (1996). Automaticity of social behavior: Direct effects of trait construct and stereotype activation on action. *Journal of Personality and Social Psychology*, 71(2), 230–244.
- [2] Batane, T. (2010). Turning to Turnitin to fight plagiarism among university students. *Educational Technology & Society*, 13(2), 1–12.
- [3] Bilic-Zulle, L., Frkovic, V., Turk, T., Azman, J., & Petroveck, M. (2008). Is there an effective approach to deterring students from plagiarizing? *Science and Engineering Ethics*, 14, 139–147.
- [4] Brehm, J. W. (1966). *A Theory of Psychological Reactance*. Academic Press.
- [5] Brooks, J. L. (2012). Counterbalancing for serial order carryover effects in experimental condition orders. *Psychological Methods*, 17(4), 600–614.
- [6] Brown, T. B., Mann, B., Ryder, N., et al. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems* 33. arXiv:2005.14165. <https://arxiv.org/abs/2005.14165>
- [7] Corneille, O., & Lush, P. (2023). Sixty years after Orne’s American Psychologist article: A conceptual framework for subjective experiences elicited by demand characteristics. *Personality and Social Psychology Review*, 27(1), 83–101. <https://doi.org/10.1177/10888683221104368>
- [8] Desaire, H., Chua, A. E., Isom, M., Jarosova, R., & Hua, D. (2023). Distinguishing academic science writing from humans or ChatGPT with over 99% accuracy using off-the-shelf machine

- learning tools. *Cell Reports Physical Science*, 4(6), 101426. <https://doi.org/10.1016/j.xcrp.2023.101426>
- [9] Jiang, F., & Hyland, K. (2025). Does ChatGPT argue like students? Bundles in argumentative essays. *Applied Linguistics*, 46, 375–391. <https://doi.org/10.1093/applin/amae052>
 - [10] Karat, C. M., Halverson, C., Horn, D., & Karat, J. (1999). Patterns of entry and correction in large vocabulary continuous speech recognition systems. *Proceedings of CHI '99*, 568–575.
 - [11] Krosnick, J. A., Holbrook, A. L., Berent, M. K., Carson, R. T., Hanemann, W. M., Kopp, R. J., ... Conaway, M. (2002). The impact of “no opinion” response options on data quality. *Public Opinion Quarterly*, 66(3), 371–403.
 - [12] Lee, K., Ippolito, D., Nystrom, A., Zhang, C., Eck, D., Callison-Burch, C., & Carlini, N. (2022). Deduplicating training data makes language models better. *Proceedings of ACL 2022*. arXiv:2107.06499. <https://arxiv.org/abs/2107.06499>
 - [13] McCambridge, J., Witton, J., & Elbourne, D. R. (2014). Systematic review of the Hawthorne effect: New concepts are needed to study research participation effects. *Journal of Clinical Epidemiology*, 67(3), 267–277.
 - [14] Mummolo, J., & Peterson, E. (2019). Demand effects in survey experiments: An empirical assessment. *American Political Science Review*, 113(2), 517–529.
 - [15] Noy, S., & Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, 381(6654), 187–192.
 - [16] Orne, M. T. (1962). On the social psychology of the psychological experiment. *American Psychologist*, 17(11), 776–783.
 - [17] Reviriego, P., Conde, J., Merino-Gomez, E., Martinez, G., & Hernandez, J. A. (2024). Playing with words: Comparing the vocabulary and lexical diversity of ChatGPT and humans. *Machine Learning with Applications*, 18, 100602. <https://doi.org/10.1016/j.mlwa.2024.100602>
 - [18] Sandler, M., Choung, H., Ross, A., & David, P. (2024). A linguistic comparison between human and ChatGPT-generated conversations. arXiv:2401.16587. <https://arxiv.org/abs/2401.16587>
 - [19] Schleimer, S., Wilkerson, D. S., & Aiken, A. (2003). Winnowing: Local algorithms for document fingerprinting. *Proceedings of the 2003 ACM SIGMOD International Conference on Management of Data*, 76–85. <https://doi.org/10.1145/872757.872770>
 - [20] Schwarz, N. (1999). Self-reports: How the questions shape the answers. *American Psychologist*, 54(2), 93–105. <https://doi.org/10.1037/0003-066X.54.2.93>
 - [21] Weingarten, E., et al. (2016). From primed concepts to action: A meta-analysis of the behavioral effects of incidentally presented words. *Psychological Bulletin*, 142(5), 472–497.
 - [22] Zenker, F., & Kyle, K. (2021). Investigating minimum text length for lexical diversity indices. *Assessing Writing*, 47, 100505.

Appendix

A. Full Assignment Table (PIDs 1 to 12)

PID	Condition	Task 1	Task 2
1	Control	For	Against
2	Test	For	Against
3	Control	Against	For
4	Test	Against	For
5	Control	For	Against
6	Test	For	Against
7	Control	Against	For
8	Test	Against	For
9	Control	For	Against
10	Test	For	Against
11	Control	Against	For
12	Test	Against	For

Table 5: Assignment table for the first 12 participant IDs, showing the repeating four-participant cycle.

B. Sensitivity Analysis

The headline result (45.87% control-as-human) was tested under several filtering conditions to check robustness:

Slice	n	Ctrl-as-human
All responses	1,999	45.87%
Drop affected docs	1,831	45.71%
Time filter only (>12s)	1,415	43.46%
Drop affected + time filter	1,298	43.45%

Table 6: Sensitivity slices. “Affected docs” are the two documents with known data-collection anomalies. The time filter removes responses faster than 12 seconds (10s forced reveal + 2s minimum). The effect strengthens under stricter filtering.

C. Full Per-Document Chosen-as-Human Rates

Doc ID	Condition	Seen	Chosen	Rate
P26_AGAINST	treatment	102	78	76.5%
P32_AGAINST	treatment	91	67	73.6%
P26_FOR	treatment	76	55	72.4%
P40_AGAINST	treatment	92	64	69.6%
P48_FOR	treatment	82	56	68.3%
P40_FOR	treatment	77	50	64.9%
P27_AGAINST	control	109	68	62.4%
P39_FOR	control	111	68	61.3%
P27_FOR	control	108	65	60.2%
P34_FOR	treatment	92	55	59.8%
<i>(middle 21 documents omitted for space)</i>				
P19_FOR	control	110	35	31.8%
P44_AGAINST	treatment	82	25	30.5%
P25_AGAINST	control	114	34	29.8%

Table 7: Top 10 and bottom 3 documents by chosen-as-human rate. Treatment documents dominate the top of the table; control documents dominate the bottom.

D. Code and Data

All analysis code, data, and the experiment platform are available at https://github.com/danieltabach/human_ai_detection. A cleaned public submission repository with the final data and analysis pipeline is available at https://github.com/danieltabach/7406_submission.